

Qlustering: Unsupervised Learning from Steady-State Currents in Open Quantum Networks

Analog quantum computation offers a route to machine learning by using controllable physical dynamics as computational resources. However, many approaches rely on task-specific protocols or observables that are difficult to access experimentally, limiting generality and implementation. Here we show that Qlustering, an unsupervised clustering framework based on steady-state quantum transport, extracts unlabeled data structure through measurable terminal currents. Data are encoded as input states of a tight-binding network governed by the GKSL master equation, and cluster assignments are inferred from correlations in steady-state output currents, avoiding full state tomography. Optimizing the Hamiltonian defines a transport-induced current-space representation in which related inputs generate similar current signatures. Benchmarks on synthetic datasets, localization, molecular QM9 subsets, and Iris show competitive performance and stability over broad dephasing strengths. These results identify terminal-current readout as a native, tomography-free mechanism for unsupervised learning in open quantum networks.

I. INTRODUCTION

Quantum transport in engineered systems provides a physically grounded route to information processing based on continuous-time dynamics. In analog quantum platforms, computation can be encoded directly in Hamiltonian evolution, scattering, interference, and open-system relaxation, allowing the dynamics of the device to perform part of the computational task [1, 2]. This perspective is particularly relevant for machine learning, where useful representations of data often emerge from high-dimensional transformations rather than from explicit symbolic rules [3, 4]. Open quantum networks are natural candidates for such transformations: an injected state propagates through a structured network, interacts with controlled dissipation, and produces measurable output observables that depend on both the input and the network Hamiltonian.

A central challenge for using quantum dynamics as a learning resource is the choice of readout. Many quantum machine-learning and analog-computing protocols rely on task-specific encodings, transient dynamical signatures, or observables that are difficult to access experimentally. Full state tomography imposes a substantial measurement overhead and generally scales poorly with system size [5, 6]. Time-resolved coherent signatures can also be fragile, since they may be strongly affected by decoherence and device imperfections [7, 8]. These considerations motivate learning protocols based on stable, experimentally accessible observables. In driven-dissipative transport networks, steady-state terminal currents provide precisely such observables: they are local transport quantities, can be measured without reconstructing the full density matrix, and naturally reflect the interplay between coherent propagation, network structure, and environmental coupling.

Recent work has shown that steady-state quantum transport networks can support supervised learning tasks in which output currents serve as the computational readout [9–11]. In these settings, an input state injected into a network is transformed by scattering, interference, and dissipation before reaching designated output terminals.

The resulting current vector acts as a transport-induced representation of the input. Related inputs can generate correlated current patterns, whereas different inputs can be routed toward distinct output signatures. This suggests a broader question: can the same transport mechanism reveal structure in unlabeled data, without learning a prescribed label map? For quantum states, this question is especially meaningful because similarity is not fixed uniquely by a classical distance, but may depend on overlap, distinguishability, localization, or measurement statistics [12]. Unsupervised learning therefore tests whether the dynamics themselves induce a useful organization of the data [13–15].

Here we introduce Qlustering, an unsupervised clustering framework based on steady-state currents in open quantum transport networks governed by the GKSL master equation. Data are encoded as input states of a tight-binding network, and the output of the computation is the vector of steady-state currents measured at terminal nodes. Cluster assignments are inferred from correlations in these current signatures, rather than from full state reconstruction. By optimizing the hopping amplitudes of the network Hamiltonian with respect to an unsupervised objective, Qlustering constructs a nonlinear current-space representation in which related inputs tend to produce similar terminal-current patterns. The central contribution is not a new microscopic transport model, nor a claim of generic superiority over classical clustering methods, but the identification of steady-state terminal currents as a native, measurable representation for label-free structure extraction in open quantum networks.

We evaluate Qlustering on controlled synthetic datasets, a localization task based on the inverse participation ratio, molecular subsets of QM9, and the Iris benchmark. These examples are chosen to test different aspects of the current-space representation: separability under controlled overlap, sensitivity to physically meaningful localization structure, recovery of geometry-linked molecular descriptors, and the effect of classical preprocessing through state normalization. We compare the method with k-means as a standard reference algo-

rithm, not as an exhaustive benchmark over all clustering methods, and use the comparison to identify the operating regime of the transport-based approach. We further examine robustness to dephasing and discuss implementation requirements in platforms with programmable couplings, input-state preparation, open-system injection and extraction, and terminal-current readout. Together, these results show that steady-state transport observables can provide a tomography-free physical primitive for unsupervised learning in open quantum networks.

II. RESULTS

A. Steady-state currents as a clustering representation

We begin by describing the physical mechanism underlying Qclustering. The goal is to assign a set of input state vectors $\Psi_{n=1}^N$, each embedded in an L -dimensional Hilbert space, into a prescribed number of clusters q . Rather than defining the clusters directly through a classical distance in the input space, Qclustering maps each input state to a vector of steady-state currents generated by an open quantum transport network. Clustering is then performed in this current space.

The network consists of L input nodes, M intermediate nodes, and q output terminals, as illustrated in Fig. 1. The input and output nodes are coupled to external reservoirs, allowing particles to be injected into the network and extracted from the output terminals. The coherent part of the dynamics is governed by a tight-binding Hamiltonian

$$\mathcal{H} = \sum_{i,j} h_{ij} c_i^\dagger c_j + \text{h.c.}, \quad (1)$$

where $c_i^\dagger$ and c_i create and annihilate a particle at node i , respectively. The hopping amplitudes h_{ij} define the trainable network parameters. This Hamiltonian form describes a broad class of quantum transport settings, including quantum-dot networks, excitonic systems, optical waveguides, programmable nanophotonic processors, and qubit networks [16–21].

The open-system dynamics are described by the Gorini–Kossakowski–Sudarshan–Lindblad master equation [22, 23],

$$\begin{aligned} \dot{\rho} &= -i[H, \rho] + \sum_k \left(V_k \rho V_k^\dagger - \frac{1}{2} \{ V_k^\dagger V_k, \rho \} \right) \\ &= -i[H, \rho] + \mathcal{L}[\rho] \end{aligned} \quad (2)$$

where ρ is the density matrix and the operators V_k describe injection, extraction, and, when included, dephasing. A data point Ψ_n is encoded through the input operator $V_{\text{in}} = \gamma_{\text{in}}^{1/2} \sum_{i=1}^L \Psi_n(i) c_i^\dagger$, where γ_{in} is the input rate and $\Psi_n(i)$ is the i th component of the input vector [17]. Extraction from output terminal r is described by

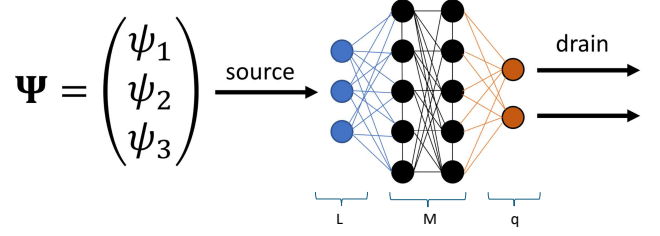


FIG. 1. Qclustering network and terminal-current readout. An input state vector Ψ of dimension L is injected into the input nodes of an open quantum transport network. The particle propagates through a tight-binding network with M intermediate nodes and is extracted through q output terminals, corresponding to the prescribed number of clusters. After the GKSL dynamics reaches the steady state, the current at each output terminal is measured. The resulting current vector defines the transport-induced representation of the input, and the state is assigned to the output terminal carrying the largest current.

$V_{\text{out},r} = \gamma_{\text{out}}^{1/2} c_r$, where γ_{out} is the output rate. Thus, each input vector is physically represented by the way it drives transport through the network.

For a fixed Hamiltonian, each input state is injected into the network and propagated until the steady-state readout regime is reached. The measured output is the terminal-current vector $\mathbf{J}[\Psi_n] = (J_1[\Psi_n], \dots, J_q[\Psi_n])$, where $J_r[\Psi_n]$ is the steady-state current extracted at output terminal r and computed as described in Methods. The cluster assignment is determined by the dominant current, $\mathcal{C}[\Psi_n] = \arg \max_r J_r[\Psi_n]$. In this way, the output currents define the representation in which clustering is performed, rather than merely providing a measurement channel.

Qclustering optimization, described in Methods, amounts to tuning the hopping amplitudes h_{ij} so that related inputs produce correlated current signatures. The optimization objective favors Hamiltonians for which each input generates a well-defined dominant terminal current, while allowing the assignment itself to change during the run. This differs from supervised current-based classification [11], where the target label associated with each input is fixed externally. Here, the grouping emerges from the transport dynamics: the optimized Hamiltonian constructs a nonlinear current-space embedding in which similar inputs tend to excite similar transport pathways and output-current patterns.

Representative optimization snapshots, including the corresponding Hamiltonian visualizations, are provided in Supplementary Fig. S1. A full movie of the optimization trajectory is provided as Supplementary Movie 1.

Operationally, Qclustering is implemented as a hybrid classical–quantum workflow. Data preparation, Hamiltonian updates, cost-function evaluation, convergence checks, consensus clustering, and validation are performed classically. The computational observable, how-

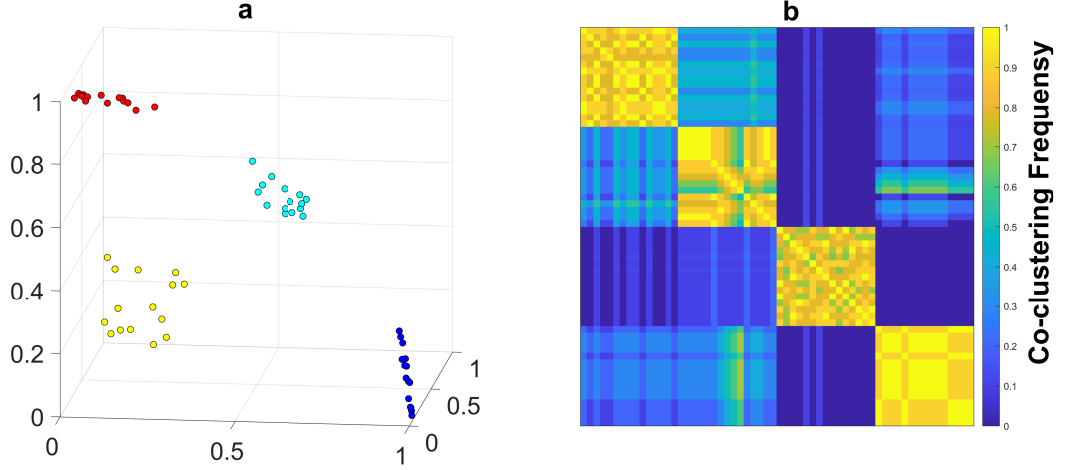


FIG. 2. **Qlustering of 3-dimensional vectors into four groups at $\omega = 0.15$.** (a) Spatial distribution of the input state vectors. (b) Consensus matrix from 10 repeated Qlustering runs. Yellow regions indicate high co-clustering frequency, while blue denotes low agreement. Each square represents a vector pair, with color intensity indicating the frequency of co-clustering (yellow: high consistency; blue: low agreement). A stable Qlustering pattern is observed, with four distinct block structures emerging in the consensus matrix, reflecting strong consistency in group assignments across runs.

ever, is generated by the open quantum network itself: each input state is injected, propagated under GKSL dynamics, and mapped to steady-state terminal currents. The details of the optimization protocol, current calculation, validation metrics, and consensus procedure are given in Methods.

B. Clustering of points in 3D space

We first ask whether the current-space representation preserves a controlled cluster structure, and how it fails when that structure is progressively destroyed. To this end, we use synthetic input states whose geometry can be tuned continuously. The inputs Ψ_n are normalized vectors in a d -dimensional Hilbert space, so that all points lie on the $(d-1)$ -dimensional unit sphere. Groups are generated around prescribed base points b_i by drawing $\Psi_n = (1 - \omega)b_i + \omega u_n$, followed by normalization, where

u_n is a random unit vector and $0 < \omega < 1$ controls the group width. For small ω , the groups are well separated; for large ω , they overlap and eventually approach a uniform distribution on the unit sphere.

We first consider a four-cluster example in three dimensions at $\omega = 0.15$. The base points are $b_1 = 0.99, 0.11, 0.11$, $b_2 = 0.11, 0.99, 0.11$, $b_3 = 0.11, 0.11, 0.99$, and $b_4 = \frac{1}{\sqrt{3}}1, 1, 1$. Figure 2 shows that the optimized transport network assigns the 60 input states to four current-space clusters that reproduce the underlying structure, yielding $\text{RI} = \text{ARI} = 1$. The consensus matrix obtained from ten repeated Qlustering runs exhibits four clear block structures, showing that the current-based assignments are stable with respect to stochastic initialization. This indicates that, when the input structure is well separated, the optimized Hamiltonian produces a reproducible terminal-current representation of the data. The full consensus clustering protocol is described in Sec. IV D

We then increase the difficulty by considering five clusters in the same three-dimensional Hilbert space and varying the width ω . The 60 data points are centered at $0, 1, 0$, $0, 0, 1$, $1, 0, 0$, $\frac{2\sqrt{3}}{9} - 1.5, 1.5, 1.5$, and $\frac{2}{\sqrt{13}}0, 1, 1.5$. The resulting scores are shown in Fig. 3. For small widths, Qlustering again achieves perfect clustering, while consensus clustering maintains high scores over a broader range of ω than individual runs. Performance decreases only when the groups approach the overlap barrier, where points begin to lie at compara-

ble distances from more than one cluster center. In the four-cluster case, effective overlap begins around $\omega \simeq 0.3$, where the scores decrease to $\text{RI} = 0.85$ and $\text{ARI} = 0.63$. Thus, in this controlled setting, the limiting factor is not the terminal-current readout itself but the loss of separability in the input structure that the transport dynamics can encode.

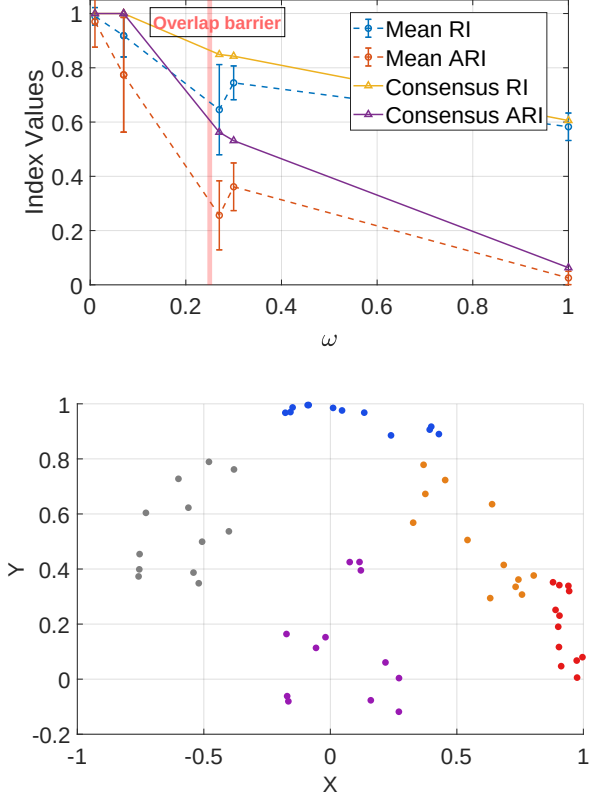


FIG. 3. **Top: Clustering scores across varying group widths, ω .** Solid-line triangles indicate the consensus clustering scores for the Rand Index (RI) and Adjusted Rand Index (ARI), while dashed-line circles represent the corresponding mean values. High clustering performance is maintained until significant overlap between groups occurs (marked by the transparent red line), after which scores decline toward random clustering levels as the distribution approaches uniformity. **Bottom: The overlap point visualized.** A 2D projection of the spatial distribution of five groups in a three-dimensional Hilbert space at $\omega = 0.25$, marking the onset of the overlap point. Colors indicate the clusters determined by the Qlustering algorithm. The groups remain largely separated, i.e. no true overlap yet, though some points already lie at approximately equal distances from their own center and that of another group. This regime marks the transition where Qlustering performance begins to decline from perfect scores toward reduced accuracy, as shown in the top figure.

C. Localization-based separation

We next ask whether Qlustering can separate data according to a physical property of the input states, rather than only according to low-dimensional geometric separation. We use the localization task studied in Ref. 11, where the relevant distinction is whether the weight of a normalized state vector is concentrated on a few sites or distributed over many sites. The degree of localization is

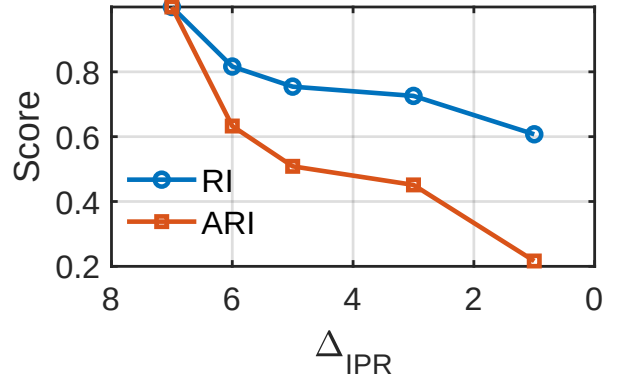


FIG. 4. **Clustering performance on the localization task.** Rand Index (RI) is shown in blue and Adjusted Rand Index (ARI) in orange on varying Δ_{IPR} . The network clusters 10-dimensional state vectors with IPR values ranging from 1 to 10. As Δ_{IPR} decreases, performance declines. This drop corresponds to the mixing of strongly localized and delocalized states, which reduces the distinctiveness of the underlying physical classes.

quantified by the inverse participation ratio [24],

$$IPR[\Psi_n] = \left(\sum_{i=1}^L |\Psi_n(i)|^4 \right)^{-1}. \quad (3)$$

With this convention, $IPR = 1$ corresponds to a fully localized state and $IPR = L$ to a uniformly extended state.

The dataset consists of 50 random state vectors with $L = 10$, separated into localized and extended groups by selecting states with low or high IPR values. We define the class separation by $\Delta_{IPR} = IPR_2^{\text{lowest}} - IPR_1^{\text{highest}}$, where IPR_1^{highest} is the largest IPR in the localized group and IPR_2^{lowest} is the smallest IPR in the extended group. Large Δ_{IPR} therefore corresponds to a clear physical separation between localized and extended states, whereas small Δ_{IPR} produces a smaller gap between the two groups, hence less separability. The task-specific current objective used for this benchmark is described in Methods.

Figure 4 shows that Qlustering maps this localization distinction into terminal-current signatures. For large Δ_{IPR} , the clustering is essentially perfect, with $RI = ARI = 1$. As Δ_{IPR} decreases, the scores fall because the localized and extended groups become less distinguishable. The decrease is not abrupt: even when the two IPR ranges overlap substantially, the Rand index remains in the range $0.7 < RI < 0.8$. These results show that the current-space representation can encode a physically meaningful property of quantum states, namely their degree of spatial localization, beyond the simple separation of compact geometric clusters.

D. QM9 dataset

To evaluate Qlustering on realistic chemical data, we used the QM9 benchmark dataset, which contains more than 130000 small molecules with quantum-chemical annotations [25, 26]. Each entry includes a SMILES representation [27], HOMO-LUMO energy gaps, temperature-dependent internal energies, and other molecular descriptors. Our main objective in this section was to assess Qlustering on a realistic, unlabeled chemical-informatics dataset.

As input to the Qlustering algorithm, and because the classical simulation of the quantum network becomes computationally expensive with increasing network dimension (see Sec. 7), we selected the 97 molecules with the smallest number of non-hydrogen atoms. This choice limits the dimensionality of the molecular representation and, consequently, the size of the simulated quantum network. The molecules were encoded using their Sorted Interatomic Distances (SID) [28, 29], computed from the 3D coordinates $\mathbf{r}_i \in \mathbb{R}^3$ of all non-hydrogen atoms:

$$\text{SID} = \text{sort} \left(\left\{ \|\mathbf{r}_i - \mathbf{r}_j\|_2 \mid i < j, Z_i, Z_j \neq 1 \right\} \right), \quad (4)$$

where Z_i denotes the atomic number of atom i . This permutation-invariant descriptor compactly encodes molecular geometry and has shown strong performance in small-molecule regression tasks [30]. Each molecule was thus represented by a fingerprint vector of length 10, corresponding to the maximum number of non-hydrogen interatomic pairs observed among the selected molecules. Shorter vectors were zero-padded to this length to ensure uniform dimensionality.

The molecular fingerprint vectors were normalized into state vectors Ψ satisfying $\sum_j |\Psi(j)|^2 = 1$. Unlike the preceding benchmark tasks, QM9 does not provide a unique ground-truth clustering partition for the present problem, which is a natural situation in unsupervised learning. Consequently, RI and ARI cannot serve here as conventional external validation metrics. The main assessment of Qlustering on QM9 therefore relies on the internal metrics described in Sec. IV: compactness, Dunn Index, silhouette, and stability.

Qlustering was first performed with $q = 2$ groups, reaching compactness = 19.87, Dunn Index = 0.765, silhouette = 0.684, and stability = 0.754. A comparison with k-means is given in Sec. I. To examine the reproducibility and internal structure of this partition, we performed 10 consecutive Qlustering runs and constructed the consensus matrix shown in Fig. 5. Although the network was constrained to produce two output groups, the matrix contains four persistent block-like substructures. This suggests that the $q = 2$ partition may combine finer structures present in the data and motivates a second calculation with $q = 4$, to which we return at the end of this section.

In addition to this primary clustering assessment, we used QM9 for a secondary analysis of the relation between the information encoded at the input and the

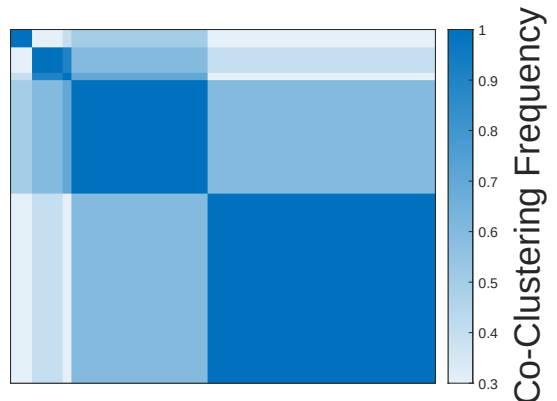


FIG. 5. **Consensus heatmap for Qlustering on a QM9 subset with $q = 2$.** Consensus values from 10 Qlustering runs on 97 QM9 molecules are shown. The approximately four-block structure suggests finer organization within the two-cluster partition and motivates the subsequent $q = 4$ analysis.

structure recovered at the output. This is not intended as an additional benchmark of clustering accuracy. Rather, it addresses a complementary question that is relevant when designing Qlustering experiments: whether a physically informed input representation can make the resulting partition preferentially sensitive to a desired class of molecular features.

To examine this input-output relation, we considered 14 molecular descriptors available in QM9. For each descriptor, its mean value within the selected subset was calculated and the molecules were divided into two groups according to whether their value was above or below this mean. RI and ARI (Eqs. (13) and (14)) were then used to quantify the agreement between each descriptor-induced partition and the $q = 2$ Qlustering partition. In this context, RI and ARI are therefore descriptor-alignment measures rather than conventional external validation scores.

The choice of SID provides a simple physical expectation for this analysis. SID is constructed directly from interatomic distances and therefore emphasizes molecular geometry. We consequently expect molecular properties that are strongly determined by geometry to show stronger alignment with the resulting Qlustering partition than properties that are only weakly related to the encoded structural information. The rotational constants RotA, RotB, and RotC provide a particularly direct test of this expectation. They are given by

$$A = \frac{h}{8\pi^2 c I_a}, \quad B = \frac{h}{8\pi^2 c I_b}, \quad C = \frac{h}{8\pi^2 c I_c}, \quad (5)$$

where I_a , I_b , and I_c are the principal moments of inertia, h is Planck's constant, and c is the speed of light. Since the principal moments of inertia are determined by

Property	RI	ARI
RotC (GHz)	0.9	0.76
RotB (GHz)	0.8	0.54
RotA (GHz)	0.71	0.34
Polarizability (a_0^3)	0.59	0.17
Internal Energy 0K (au)	0.58	0.15
Internal Energy 298K (au)	0.58	0.15
Enthalpy 298K (au)	0.58	0.15
Gibbs Free Energy 298K (au)	0.58	0.15
HOMO (au)	0.57	0.09
$\langle R^2 \rangle (a_0^2)$	0.56	0.12
Heat Capacity ($\text{cal mol}^{-1} \text{K}^{-1}$)	0.55	0.1
LUMO (au)	0.51	-0.03
ZPVE (au)	0.51	0.02
Dipole (Debye)	0.5	-0.01
HOMO-LUMO gap (au)	0.5	0

FIG. 6. **Descriptor alignment for a single Qlustering run on the QM9 subset ($q = 2$).** RI and ARI quantify the agreement between the Qlustering partition and binary partitions defined from 14 molecular descriptors. The geometry-dependent rotational constants RotA, RotB, and RotC show the strongest alignment, with RotC reaching RI = 0.90 and ARI = 0.76.

the spatial distribution of atomic masses, the rotational constants provide a direct connection between the molecular geometry encoded by SID and measurable molecular properties.

The descriptor-alignment scores for RotA descriptor of RI = 0.66 and ARI = 0.23, RotB descriptor of RI = 0.65 and ARI = 0.23, and RotC descriptor scores of RI = 0.71 and ARI = 0.32. A single $q = 2$ Qlustering run are shown in Fig. 6. Among the 14 descriptors considered, the strongest alignment was obtained for the rotational constants. RotC reached RI = 0.90 and ARI = 0.76, while RotA and RotB also yielded relatively high scores, with RI > 0.70 and ARI > 0.30.

This secondary analysis highlights the role of input design in Qlustering. The strong alignment with the rotational constants indicates that the geometrical information emphasized by the SID representation remains reflected in the output partition, without treating these descriptors as true cluster labels. More generally, this suggests that physically informed input representations can be chosen to emphasize features relevant to the intended clustering objective. In the present case, encoding interatomic geometry leads to clusters that preferentially align with geometry-dependent molecular proper-

ties, providing a practical guideline for experiment planning.

Returning to the main QM9 clustering analysis, the finer substructure suggested by the consensus heatmap in Fig. 5 motivated an additional calculation with $q = 4$. Ten independent Qlustering runs were performed. The resulting internal scores were compactness = 0.865, Dunn Index = 2.4251, silhouette = 0.9776, and stability = 0.80. All four metrics therefore change in the direction associated with a more compact, separated, and reproducible partition relative to the $q = 2$ result, supporting the presence of finer structure in this subset. A comparison with k-means (Table I) shows that for $q = 4$, Qlustering achieves lower compactness and higher Dunn Index and silhouette scores, whereas k-means exhibits higher run-to-run stability.

E. Iris dataset

The Iris dataset, introduced by Ronald A. Fisher in 1936 [31] and obtained from the UCI Machine Learning Repository [32], is a classic benchmark in pattern recognition and unsupervised learning. It contains 150 samples from three Iris species (*Iris setosa*, *Iris versicolor*, and *Iris virginica*), each described by four features: sepal length, sepal width, petal length, and petal width. Although the dataset includes labeled classes, it is widely used in clustering tasks to assess an algorithm’s ability to recover natural groupings without prior knowledge. Its well-structured yet partially overlapping class distributions make it a useful testbed for comparing clustering methods [33–36].

The dataset was clustered using Qlustering, with internal and external evaluation scores measured over 10 consecutive runs. We evaluated both the original four-feature dataset and a reduced three-feature representation obtained by removing sepal width. Feature selection and subspace analysis are established procedures in Iris dataset clustering, since different feature subsets can expose different cluster structures [? ?]. Dy and Brodley demonstrated on the dataset that its cluster structure can be recovered using only a subset of the original features [?]. The reduced representation is therefore included as an additional controlled test rather than as a replacement for the full benchmark. A 4-2-3 architecture was used for the full representation, with the input layer reduced to an overall 3-2-3 network accordingly for the three-feature case. The full representation exhibits a more overlapping and less convex cluster geometry, whereas removing sepal width produces a more convex and separable representation.

Using all four features, Qlustering achieved RI = 0.77, ARI = 0.56, and internal metrics of stability = 0.72, compactness (CP) = 458.3, Dunn Validity Index (DVI) = 0.038, and silhouette score = 0.14 (all mean values over 10 runs). For the reduced three-feature representation, the external scores increased to RI = 0.92 and ARI =

0.82, with internal metrics of stability = 0.60, CP = 2.5, DVI = 0.022, and silhouette score = 0.38.

The difference between the two representations is consistent with the overlap barrier observed in Fig. 3: Qlustering performance decreases for the more overlapping and non-convex full-feature geometry, whereas the more convex and separable reduced geometry yields substantially better external clustering performance. These results indicate that data with convex and well-separated cluster structures are well suited to Qlustering, while the full-feature result illustrates its sensitivity to overlap and non-convexity. This difficulty may be further enhanced by normalization-induced skewing, as discussed in Appendix ??; this effect can be mitigated through the pre-processing and encoding strategies described there and in the main text.

F. Comparison with k-means

We use k-means as a standard reference method to identify the operating regime of Qlustering, rather than as an exhaustive benchmark over clustering algorithms. This comparison is useful because k-means is computationally efficient, widely used, and directly optimizes compactness in Euclidean feature space [37, 38]. It therefore provides a natural baseline for asking when a transport-induced current-space representation behaves similarly to a conventional geometric clustering method, and when it provides a distinct organization of the data.

The comparison is summarized in Table I and Fig. 7. For compact synthetic clusters, k-means performs very well, as expected for a centroid-based method. In particular, it remains strong at $\omega = 0.3$, where Qlustering begins to lose accuracy as the input groups approach the overlap barrier. This behavior is consistent with the interpretation of the synthetic benchmark in Sec. II B: Qlustering is effective when the optimized current-space representation can preserve separability, but its performance decreases when the input structure becomes too mixed to induce distinct terminal-current signatures.

The localization and QM9 results show a different regime. In the localization task, k-means performs close to random on the external scores, whereas Qlustering remains sensitive to the localized–extended distinction until the IPR ranges become nearly mixed. This indicates that the terminal-current embedding can capture a physically meaningful property of the state vectors that is not well represented by simple centroid separation in the original feature space. In QM9, k-means gives strong internal scores but weaker agreement with the geometry-linked molecular descriptors used for external validation. By contrast, Qlustering shows its strongest agreement with rotational constants, especially RotC, consistent with the fact that both the SID input representation and rotational constants depend on molecular geometry. For Iris, the comparison emphasizes a preprocessing constraint: with all four normalized features, k-means performs bet-

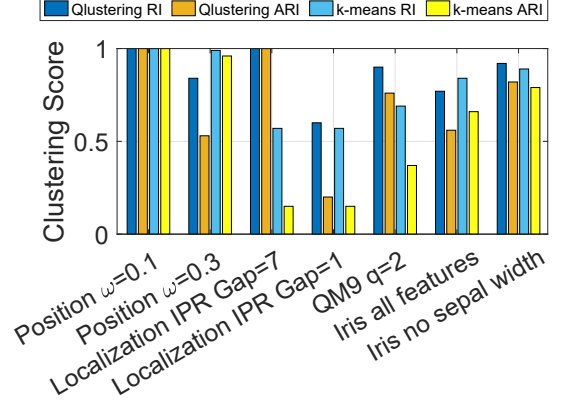


FIG. 7. **Operating-regime comparison between Qlustering and k-means.** Rand Index (RI) and Adjusted Rand Index (ARI) scores are shown for both algorithms across the main benchmarks: positional clustering at $\omega = 0.1$ and $\omega = 0.3$, localization with $\Delta_{\text{IPR}} = 7$ and $\Delta_{\text{IPR}} = 1$, QM9 with $q = 2$, and Iris with and without sepal width. The comparison highlights complementary regimes: k-means is strong for compact geometric clusters, whereas Qlustering is advantageous for the localization task and captures geometry-linked molecular correlations in QM9.

ter, whereas removing sepal width improves the Qlustering scores substantially. As discussed in the Supplementary Information, this normalization sensitivity may be mitigated by alternative preprocessing strategies. Thus, the comparison supports the competitive behavior of Qlustering while also defining its operating regime: the method is most useful for structured high-dimensional inputs and data with physically meaningful correlations, whereas some classical datasets may require additional preprocessing before they are naturally represented as quantum-state inputs.

For a physical implementation, the relevant operations are input preparation, propagation to the readout regime, terminal-current measurement, and Hamiltonian reconfiguration during training. Let i be the number of training iterations, P the number of candidate perturbations evaluated per iteration, N the number of input states, L the input-state dimension, q the number of output terminals, t_{st} the relaxation time to the readout regime, and h the number of nonzero hopping amplitudes in $\mathcal{H}$. The resulting training cost is

$$\begin{aligned} T_{\text{iter}} &= \mathcal{O}(P(NL + Nt_{\text{st}}^{CF} + Nq + h + 1)) \\ &\simeq \mathcal{O}(P(NL + Nt_{\text{st}}^{CF} + Nq + h)). \end{aligned}$$

and the total training complexity over i iterations is

$$\mathcal{T}_{\text{train}} = \mathcal{O}(iP(NL + Nt_{\text{st}}^{CF} + Nq + h)). \quad (6)$$

TABLE I. K-Means Clustering Performance Scores

Score	Position		Localization		QM9		Iris
	$\omega = 0.1$	$\omega = 0.3$	$\Delta_{IPR} = 7$	$\Delta_{IPR} = 1$	$q = 2$	$q = 4$	
External Scores							
RI	1	0.987	0.571	0.571	0.693	---	0.843
ARI	1	0.957	0.150	0.150	0.373	---	0.659
Internal Scores							
CP	0.482	8.31	14.109	9.73	15.434	1.198	91.905
DVI	3.337	0.132	0.415	0.257	0.674	2.188	0.095
Silhouette	0.981	0.630	0.417	0.345	0.697	0.716	0.716
Stability	1.00	0.790	0.846	0.750	0.851	0.940	0.851

G. Computational cost and physical implementation

The computational complexity of Qclustering depends on whether the protocol is classically simulated or physically implemented. In a classical simulation, the main cost is evaluating the steady-state current for each input and Hamiltonian. This requires solving for the null space of a $d^2 \times d^2$ Liouvillian matrix, where d is the total number of network nodes. A naive dense treatment therefore scales as $\mathcal{O}((d^2)^3) = \mathcal{O}(d^6)$. This cost reflects the overhead of simulating open-system dynamics on a classical computer and should not be identified with the runtime of the physical protocol.

In a physical analog implementation, the relevant complexity is instead determined by propagation and relaxation, input preparation, terminal-current measurement, Hamiltonian reconfiguration, and classical-to-quantum communication and control. We assume parallel control of current readout and qubit preparation, or the corresponding input-state preparation on other platforms. Let i be the number of training iterations, P the number of candidate Hamiltonian perturbations evaluated per iteration, and $P_{\parallel} \leq P$ the number evaluated simultaneously. The physical runtime can then be written as

$$t_{\text{iter}} = \mathcal{O}\left[\left[\frac{P}{P_{\parallel}}\right] (t_{\text{conf}} + N [t_{\text{prep}} + t_{\text{read}} + t_{\text{st}}^{CF}])\right],$$

$$t_{\text{train}} = \mathcal{O}(i t_{\text{iter}}). \quad (7)$$

Here, t_{prep} is the time required for input preparation, t_{conf} is the time required for Hamiltonian reconfiguration, and t_{read} is the readout time. N is the number of input states, and t_{st}^{CF} is the relaxation time to the cost-function steady-state regime, defined as the time required for the terminal currents to become stable and sufficiently separated such that their relative ordering no longer changes. The dependence on circuit size, including L , q , and h , is implicit in t_{st}^{CF} through the Liouvillian gap of the network and is assessed below for different tasks. Since the Hamiltonian is configured once for each candidate perturbation and subsequently used for all N input states, t_{conf} does not acquire a factor of N .

Under parallel control, preparation, reconfiguration, and readout operations increase the required hardware

resources with network size rather than necessarily increasing the elapsed runtime. Likewise, evaluating all P candidate perturbations in parallel increases the hardware requirement rather than the physical runtime. For fixed relaxation and control times and a fixed degree of candidate parallelization, the physical runtime is therefore linear in the dataset size N and in the number of training iterations i .

For comparison, the standard Lloyd implementation of k-means scales as $\mathcal{O}(NKDi)$ for N data points, K clusters, feature dimension D , and i iterations [37, 38]. Gaussian-mixture models trained by expectation-maximization scale as $\mathcal{O}(i(NKD^2 + KD^3))$ for full covariance matrices, or as $\mathcal{O}(NKDi)$ for diagonal covariances [39]. DBSCAN typically scales as $\mathcal{O}(N \log N)$ when efficient spatial indexing is available in low-dimensional data, but can reach $\mathcal{O}(N^2)$ in the worst case [40]. Dense spectral clustering is usually limited by construction and eigendecomposition of an $N \times N$ affinity matrix, giving $\mathcal{O}(N^3)$ time and $\mathcal{O}(N^2)$ memory [41]. These classical operation counts are not directly equivalent to the physical elapsed-time estimate in Eq. (7).

Effective-model hardware requirements.

At the level of the effective model, a Qclustering device requires $d = L + M + q$ network nodes, comprising L input nodes, M intermediate nodes, and q monitored output nodes (Fig. 1). The intermediate nodes are mutually interconnected, and each is coupled to all input and output nodes, whereas no direct couplings exist within or between the input and output layers. The number of independently programmable hopping amplitudes (pairwise couplings) is therefore

$$h = LM + Mq + \frac{M(M-1)}{2}.$$

For example, the 3-2-5 network used in the position task contains $d = 10$ nodes and $h = 17$ pairwise couplings, whereas the 10-3-2 QM9 network contains $d = 15$ nodes and $h = 39$.

A physical Qclustering platform must support programmable couplings, controllable input-state preparation, open-system injection and extraction, and terminal-current readout. The central practical feature is that the measured quantities are local terminal currents, avoiding full density-matrix reconstruction and detailed time-

resolved tomography. The corresponding physical resources and device layout depend on the implementation platform.

Superconducting quantum computers.

One possible implementation uses superconducting quantum computers (QC) in an analog rather than circuit-based approach. A blueprint for an analog superconducting implementation is provided in Ref. [42].

A transmon-based implementation requires, at minimum, d transmon modes, L independently addressable microwave drive channels for input-state preparation, h independently tunable couplers, and q readout resonators for dispersive measurement of the output nodes [43]. If all coupling strengths h_{ij} are reconfigured independently and in parallel, a flux-tunable coupler architecture requires h dedicated flux-bias control channels. Readout does not require q independent microwave feedlines, as frequency-multiplexed readout can be employed across shared lines. Implementing the source-sink dynamics of the GKSL master equation further requires reservoir engineering, such as driven auxiliary lossy modes for dissipation and state injection.

Programmable photonic processors.

Photonic transport architectures provide a natural setting for these requirements. Integrated photonic systems have demonstrated related random-walk transport [44], while programmable photonic processors provide controllable input preparation, network connectivity, and terminal readout [45?]. In addition, LPU processors can provide a similar programmable platform.

In a programmable photonic processor, the network is represented by spatial optical modes interconnected through a reconfigurable waveguide mesh. The corresponding hardware requirements are d optical modes, L independently addressable optical input ports, a mesh containing sufficient programmable unit cells (PUCs) to synthesize the h required couplings, and q monitored output ports or integrated photodetectors [?]. Each PUC is typically implemented as a 2×2 Mach-Zehnder interferometer controlled by thermo-optic phase shifters to provide programmable coupling ratios and phase shifts. The precise number of required PUCs depends on the mesh topology (e.g., Clements or Reck architecture) and may exceed h because of routing overhead. The required hopping amplitudes may be realized through nearest-neighbor links, programmable meshes, or effective nonlocal couplings, making the detailed device layout platform-dependent. Reproducing the source-sink dynamics of the GKSL model additionally requires controlled optical loss, for example through variable optical attenuators, together with optical pumping or amplification.

Parallel evaluation of P candidate Hamiltonians requires P independent spatial realizations of these platform-specific resources. Sequential reuse of a single device exchanges this hardware footprint for increased total runtime.

Relaxation, control, and readout times.

Besides readout time, the dominant system-dependent timescale in Eq. (7) is $t_{\text{st}}^{\text{CF}}$. It depends on the Liouvillian spectral gap, the reservoir couplings, the network dimensions, and the input state. It may therefore increase in larger or slowly relaxing networks. However, convergence to a stable readout regime is sufficient rather than convergence to the full steady state.

To estimate this timescale, we evaluated the ten trained Hamiltonians used for consensus clustering in the synthetic position and QM9 tasks. For the 3-2-5 position networks, $t_{\text{st}}^{\text{CF}}$ ranged from 55.1 to 332 ns, with a median of 77.6 ns and eight of the ten networks reaching the readout regime within 100 ns. For the 10-3-2 QM9 networks, nine stable Hamiltonians yielded $t_{\text{st}}^{\text{CF}}$ between 23.9 and 799 ns, with a median of 138 ns. One additional Hamiltonian required $4.896 \mu\text{s}$ and produced an unstable steady-state. Such solutions can be rejected by imposing a minimum current-separation criterion during training.

The remaining contributions in Eq. (7) are t_{conf} , t_{prep} , and t_{read} . We define $\tau_0 = t_{\text{conf}} + N(t_{\text{prep}} + t_{\text{read}})$. These quantities are system-dependent and are estimated separately for photonic computational platforms and superconducting QCs.

For superconducting quantum hardware, representative operation times are in the nanosecond-to-microsecond regime. IBM Quantum Platform documentation reports single-qubit gate durations of order tens of nanoseconds and readout times of order $1 \mu\text{s}$ [?]. A separate Sycamore-based analog superconducting implementation reports qubit- and coupler-ramp times of approximately 100 ns for setting the target Hamiltonian parameters [42]. Since readout is the slowest of these operations and also increases linearly with N , the characteristic overhead is estimated as

$$\tau_0 \sim N \cdot t_{\text{read}} \sim \mathcal{O}(N \cdot 1 \mu\text{s}),$$

with the precise value depending on the number of input data points, N , the hardware architecture, and the control protocol.

For photonic computational platforms, the hardware overhead is likewise dominated by configuration and readout. In a recent MZI-based programmable photonic processor, the combined set-and-readout time was measured as 15–90 ms [?]. Thus, for present experimentally demonstrated photonic implementations, a representative estimate is

$$\tau_0 \sim N \cdot t_{\text{read}} \sim \mathcal{O}(N \cdot 10 \text{ ms}).$$

III. DISCUSSION

We have shown that steady-state quantum transport can provide a physical mechanism for unsupervised structure extraction. In Qlustering, each input state is injected into an open quantum network and mapped to a vector of terminal currents generated by GKSL dynamics. The clustering representation is therefore not

obtained from full state reconstruction or from a prescribed sequence of gates, but from a directly measurable transport observable. By optimizing the network Hamiltonian, the method constructs a nonlinear current-space embedding in which related inputs tend to produce correlated output-current signatures. This extends current-based quantum transport networks from supervised classification to label-free organization of data.

The examples studied here identify how this current-space representation behaves. In controlled synthetic data, Qlustering gives stable assignments when the input groups are well separated and fails gradually as overlap destroys the structure that can be encoded by the transport dynamics. In the localization task, terminal currents distinguish localized from extended states, showing that the method can capture a physically meaningful property of quantum states rather than only compact geometric clusters. In QM9, the strongest external agreement occurs with rotational constants, especially RotC, consistent with the fact that both the SID input representation and rotational constants depend on molecular geometry. The Iris example highlights a complementary point: because classical data must be normalized into state vectors, preprocessing is part of the physical encoding and can either obscure or enhance separability.

The comparison with k-means should be interpreted in this operating-regime sense. K-means is a strong and efficient compactness-based method, and it performs particularly well for compact convex clusters. Qlustering is not intended to replace such methods in their natural regime, nor do the present results establish generic superiority over classical clustering. Instead, the comparison shows where a transport-induced representation can provide a distinct organization of the data. This is most evident in the localization task and in the geometry-linked molecular correlations in QM9, where the terminal-current embedding captures structure that is not simply equivalent to Euclidean compactness. The strong run-to-run stability observed in several examples also makes the method well suited to consensus clustering, which extracts reproducible pairwise structure across stochastic Hamiltonian initializations.

The present study also clarifies the main limitations. The datasets considered here are relatively small, primarily because the results were obtained by classical simulation of the open-system dynamics. In such simulations, solving for the steady state of the Liouvillian can become costly as the network size increases. A physical implementation would have a different cost structure, governed by input preparation, propagation to the readout regime, terminal-current measurement, and Hamiltonian reconfiguration. The relaxation time to the readout regime is therefore a key system-dependent parameter. Performance also depends on the separability of the induced current-space representation: strong overlap, unfavorable normalization, or an encoding that does not expose the relevant structure can reduce clustering quality.

From an implementation perspective, Qlustering is

naturally matched to analog transport platforms. The required ingredients are programmable couplings, state injection, open-system extraction, and local terminal-current readout. These requirements are most directly aligned with photonic and other transport-based architectures, where output intensities or fluxes can serve as the measured currents. The dephasing analysis supports this picture by showing that Qlustering remains effective over a broad range of dephasing strengths. The method therefore does not rely on a narrowly coherent transient response, but can operate through steady-state observables in noisy open-system conditions. Realization on gate-based hardware may be possible through synthesized GKSL dynamics, but the correspondence is less direct than in native transport platforms.

Future work should address larger datasets, automated selection of the number of clusters, preprocessing strategies adapted to normalized state-vector encodings, and optimization methods that reduce the number of Hamiltonian updates required during training. It will also be important to determine whether useful readout can be obtained before strict convergence to the steady state, which could reduce the physical runtime. More broadly, the results position steady-state terminal-current readout as a concrete physical primitive for unsupervised learning in open quantum networks.

IV. METHODS

A. Current from the Lindblad equation

We evaluate currents through a steady-state solution of the Lindblad equation. [17, 46–49] We consider a system governed by a general tight-binding Hamiltonian, with dynamics described by the Lindblad master equation:

$$\begin{aligned}\dot{\rho} &= -i[H, \rho] + \sum_k \left(V_k \rho V_k^\dagger - \frac{1}{2} \{V_k^\dagger V_k, \rho\} \right) \\ &= -i[H, \rho] + \mathcal{L}[\rho]\end{aligned}$$

The calculation is restricted to the single-exciton manifold, which suffices for capturing the relevant physics, as established in prior work [17].

We solve for the steady-state density matrix ρ_s , defined by $\dot{\rho}_s = 0$. The current J_i at site i is given by the continuity equation,

$$J_i = \frac{d\langle n_i \rangle}{dt} = \frac{d}{dt} \text{Tr}(\hat{n}_i \rho) . \quad (8)$$

where n_i is the number density in site i . Taking the time derivative inside the trace yields:

$$J_i = \text{Tr}(\dot{\hat{n}}_i \rho_s + \hat{n}_i \dot{\rho}_s) . \quad (9)$$

At steady state the expression simplifies. For exit sites, where $\dot{\hat{n}}_i = -i[\hat{n}_i, \mathcal{H}]$, the total current conservation implies:

$$J_{ext} = \text{Tr}(\hat{n}_{ext} (-i[\mathcal{H}, \rho_s] + \mathcal{L}_{ext}[\rho_s])) = 0 . \quad (10)$$

Due to current conservation, at steady state the current outside the network equals in magnitude to the current inside, and opposite in sign. Hence the zero net current. Throughout the paper, however, J_{out} denotes the positive extracted terminal current, or the output current, rather than the net balance. It is defined by the dissipative extraction contribution and is therefore not constrained to vanish:

$$J_{out} = \text{Tr}(\hat{n}_{ext} \mathcal{L}_{ext}[\rho_s]) . \quad (11)$$

B. Qlustering optimization protocol

The Qlustering optimization takes as input a set of normalized state vectors $\mathcal{D} = \{\Psi_1, \Psi_2, \dots, \Psi_N\}$, a prescribed number of clusters q , and an initial Hamiltonian $\mathcal{H}$. For a given Hamiltonian, each input state is injected into the network, propagated under the GKSL dynamics, and mapped to a steady-state current vector $\mathbf{J}[\Psi_n] = (J_1[\Psi_n], \dots, J_q[\Psi_n])$. The instantaneous assignment of Ψ_n is determined by the output terminal carrying the largest current, $i_{\max}(n) = \arg \max_r J_r[\Psi_n]$.

The optimization uses a dynamically updated unsupervised cost function. For each input state, we define a one-hot vector I_n with components $I_n(r) = \delta_{r, i_{\max}(n)}$. The cost function is

$$CF(\mathbf{J}, N) = \sum_{n=1}^N |I_n - \mathbf{J}[\Psi_n]|^2 . \quad (12)$$

Because $i_{\max}(n)$ depends on the currents produced by the current Hamiltonian, the vectors I_n are recomputed whenever $\mathcal{H}$ is changed. The cost function is therefore dynamical: it does not compare the output currents with externally supplied labels, but selects Hamiltonians that sharpen the current response by favoring a dominant output terminal for each input. This distinguishes Qlustering from the supervised current-based classification protocol of Ref. [11], where the target labels remain fixed throughout training.

In the simulations reported here, the Hamiltonian is initialized randomly. At each iteration, one hopping amplitude h_{ij} is selected and perturbed to generate a candidate Hamiltonian $\mathcal{H}^{(2)}$. The input states are then propagated through the candidate network, the steady-state current vectors are recalculated, and the cost function in Eq. 12 is evaluated. If the candidate Hamiltonian lowers the cost, it is accepted; otherwise, the previous Hamiltonian is retained. This procedure is repeated until convergence or until a predefined number of iterations is reached. To accelerate convergence, several candidate perturbations can be evaluated in parallel at each step, and the update producing the lowest cost can be selected.

The complete optimization loop is:

Qlustering optimization

- 1: Initialize the Hamiltonian $\mathcal{H}$ randomly.
- 2: **for** each input state Ψ_n **do**
- 3: Inject Ψ_n into the input nodes and propagate under GKSL dynamics.
- 4: Compute the steady-state output-current vector $\mathbf{J}[\Psi_n]$.
- 5: Assign $i_{\max}(n) = \arg \max_r J_r[\Psi_n]$.
- 6: **end for**
- 7: Construct the one-hot vectors $I_n(r) = \delta_{r, i_{\max}(n)}$ and evaluate $CF(\mathbf{J}, N)$.
- 8: Perturb one or more hopping amplitudes h_{ij} to generate a candidate Hamiltonian.
- 9: Recompute the current vectors and cost function.
- 10: Accept the candidate Hamiltonian if the cost decreases; otherwise retain the previous Hamiltonian.
- 11: Repeat until convergence or until the maximum number of iterations is reached.

Representative snapshots of the optimization protocol are shown in Supplementary Fig. S1, where selected iterations are displayed together with the corresponding Hamiltonian visualizations. The full optimization trajectory is provided as Supplementary Movie 1.

C. External and internal validation metrics

a. Rand index (RI) and Adjusted Rand index (ARI) scores The RI measures the proportion of pairwise agreements, whether two points are correctly grouped or separated, and is defined as [50, 51]:

$$\text{RI} = \frac{TP + TN}{TP + TN + FP + FN} \quad (13)$$

where TP and TN denote true positives and true negatives, and FP , FN are false positives and false negatives. RI values range from 0 (complete disagreement) to 1 (perfect agreement).

However, RI does not account for agreement due to chance; high RI values may still occur in poorly performing models. To address this, we use the ARI [50]:

$$\text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{\max(\text{RI}) - \mathbb{E}[\text{RI}]} \quad (14)$$

where $\mathbb{E}[\text{RI}]$ is the expected RI of a random model. ARI ranges from -1 to 1 , with 0 indicating agreement by chance and 1 for a perfect match. Negative values mean the clustering is worse than random.

b. Compactness (CP) Compactness measures the within-cluster dispersion by computing the sum of squared distances between data points and their respective cluster centroids [37, 38]:

$$\text{CP} = \sum_{i=1}^N \|\mathbf{x}_i - \mathbf{c}_{\text{label}(i)}\|^2$$

where $\mathbf{c}_{\text{label}(i)}$ denotes the centroid of the cluster to which point $\mathbf{x}_i$ is assigned. Lower CP values indicate tighter, more cohesive clusters.

c. Dunn Validity Index (DVI) The Dunn Index evaluates the ratio between the minimum inter-cluster distance and the maximum intra-cluster diameter [52, 53]:

$$\text{DVI} = \frac{\min_{i \neq j} \delta(C_i, C_j)}{\max_k \Delta(C_k)}$$

where:

- $\delta(C_i, C_j)$ is the minimum pairwise distance between points in clusters C_i and C_j ,
- $\Delta(C_k)$ is the maximum distance between any two points within cluster C_k .

Higher DVI values indicate well-separated and compact clusters.

d. Silhouette Score The silhouette score captures how similar an object is to its own cluster compared to other clusters. For each point $\mathbf{x}_i$, the silhouette value $s(i)$ is defined as [54, 55]:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

where:

- $a(i)$ is the average intra-cluster distance (cohesion),
- $b(i)$ is the lowest average inter-cluster distance to any other cluster (separation).

The overall silhouette score is the mean of all $s(i)$. Values close to 1 indicate good clustering; values near 0 suggest overlapping clusters.

e. Stability via Label Alignment To assess the stability of clustering across multiple runs, we align cluster labels using the Hungarian matching algorithm applied to pairwise confusion matrices. The stability metric is defined as the average alignment accuracy over all pairwise label permutations [38]:

$$\text{Stability} = \frac{2}{R(R-1)} \sum_{i < j} \text{Match}(R_i, R_j)$$

where R is the number of clustering repetitions, and $\text{Match}(R_i, R_j)$ is the fraction of matching labels between runs i and j after optimal alignment.

D. Consensus clustering procedure

To ensure robust and stable group identification in unlabeled data, we employed **consensus clustering** following the protocol of Monti et al. [56].

Step 1: Repeated clustering. We ran the clustering algorithm $R = 10$ times on the same dataset using random initializations. Each run produced a partition P_r , where $r \in \{1, 2, \dots, R\}$.

Step 2: Consensus matrix construction. An $n \times n$ consensus matrix C was constructed as:

$$C_{ij} = \frac{1}{R} \sum_{r=1}^R \mathbb{I}(x_i \text{ and } x_j \text{ belong to the same cluster in } P_r)$$

where $\mathbb{I}$ is the indicator function, and n is the number of data points.

Step 3: Final clustering. We computed a distance matrix $D = 1 - C$ and applied *hierarchical clustering with average linkage* (UPGMA) to assign the final cluster labels.

Rationale. Averaging performance metrics (e.g., RI, ARI) over multiple runs does not capture clustering stability. By contrast, consensus clustering records how frequently pairs of samples are assigned to the same cluster across repeated runs, thereby yielding a more reliable final grouping and providing both a visual and a quantitative measure of stability. In the present framework, this procedure is particularly useful because Qlustering exhibits relatively high run-to-run stability, so repeated runs tend to preserve a common underlying structure despite differences in initialization. Consensus clustering therefore serves not only as a generic post-processing step, but as a practical way to extract that shared structure and improve the robustness of the final assignment.

The underlying codes and training/validation datasets for this study are available in the code availability section.

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